Prabhat Tiwari

GitHub | LeetCode | Portfolio | geeksforgeeks

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Software Engineer Intern

Final-year Computer Science student skilled in Python, and JavaScript, with strong data structures and algorithms fundamentals. Experienced in building real-time backend systems using FastAPI, WebSockets, and Node.js, along with REST APIs and databases such as PostgreSQL and MongoDB. Built complete AI/ML and blockchain projects independently, from system design through implementation and testing.

Education

Galgotias College Of Engineering And Technology — Greater Noida, India
Bachelor of Technology - Computer Science and Engineering — *2023 – Present*

Sunbeam School — Ballia, India
Intermediate - PCM — *2022*

Technical Skills

Languages	:	Python, Java, JavaScript/TypeScript, C, C++, SQL, Solidity
Backend/Infra	:	FastAPI, WebSockets, Node.js, Express.js, PostgreSQL, Redis, MongoDB, Docker
Frontend/Tools	:	Git/GitHub, ReactJS, Redux, NextJS, VS Code, Linux, DataGrip, IntelliJ IDEA
AI/ML	:	PyTorch, PyTorch Geometric, HuggingFace Transformers, LangChain, FAISS, XGBoost, scikit-learn
Blockchain	:	Web3.py, Hardhat, IPFS

Projects

VeritasChain — *Decentralized AI Fact-Checking Pipeline* — Source Code

- Built a FastAPI + WebSocket backend streaming real-time claim submissions and verdicts to a React/TypeScript dashboard — the same low-latency, event-driven backend pattern behind real-time communications platforms.
- Designed an efficient claim-processing core — Trie/heap-based deduplication feeding a LangChain + FAISS RAG verdict-generation module and a PyTorch Geometric GNN for dynamic source-trust scoring — fully integrated end-to-end.
- Enforced structured, schema-validated service outputs (verdict, confidence, citations) via Pydantic, and built a Solidity + Web3.py bridge for tamper-proof, IPFS-backed verdict storage.

VolatilityForge — *Quantitative Market Volatility Forecasting Platform* — Source Code

- Built a quantitative research pipeline using Python, pandas, and yfinance to process and align SPY and VIX market data, engineer volatility features, and construct forward-looking realized-variance targets.
- Implemented and evaluated EWMA, GARCH, GJR-GARCH, HAR-RV, and LightGBM models for 1-day and 5-day volatility forecasting using leakage-aware expanding-window walk-forward validation, with QLIKE and MSE metrics.
- Developed statistical model comparisons, volatility-regime analysis, and transaction-cost-aware volatility-targeting backtests; built a Streamlit dashboard to explore forecast performance, strategy returns, drawdowns, and sensitivity to trading costs.

Certifications

- Complete web development course
- The Complete Data Structures and Algorithms Course in Python
- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents
- Machine Learning A-Z [2026]: ML, DL, AI with AWS, Python
- Blockchain A-Z: Build a Blockchain, Crypto, & Smart Contract